

Bisection Method

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1 Introduction

The Bisection Method is a numerical root-finding algorithm used to find a solution of the nonlinear equation

$$f(x) = 0.$$

It is one of the simplest and most reliable root-finding methods because it only requires the function to be continuous on an interval where the function values at the endpoints have opposite signs.

2 Mathematical Foundation

2.1 Intermediate Value Theorem

If f is continuous on a closed interval $[a, b]$, and c is any number between $f(a)$ and $f(b)$ inclusive, then there is at least one number x in the closed interval such that $f(x) = c$.

2.2 Bisection Method

The Bisection Method is based on the Intermediate Value Theorem. Let f be continuous on the interval $[a, b]$. If $f(a)f(b) < 0$, then there exists at least one point $c \in [a, b]$ such that $f(c) = 0$.

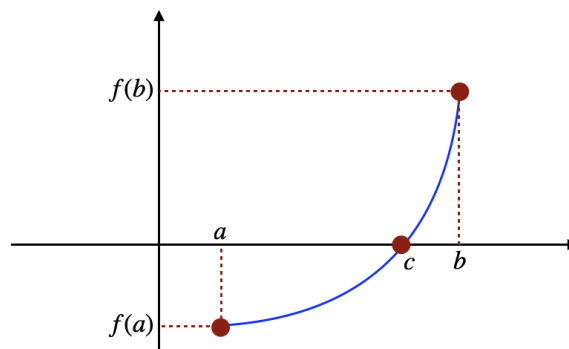


Figure 1: Bisection Method

3 Algorithm Flow

At every iteration, the current interval is divided into two equal sub-intervals. The sub-interval that still contains a sign change is guaranteed to contain at least one root. Therefore, the interval containing the root becomes smaller after every iteration.

Algorithm 1 Bisection Method

Input: Function $f(x)$, Interval $[a, b]$, Tolerance ϵ , Maximum iterations N

Output: Approximate root x

```
if  $f(a)f(b) \geq 0$  then
    return failure
end if
for  $k = 1$  to  $N$  do
     $m = a + \frac{b-a}{2}$ 
    if  $|f(m)| < \epsilon$  then
        return  $m$ 
    end if
    if  $f(a)f(m) < 0$  then
         $b = m$ 
    else
         $a = m$ 
    end if
end for
return failure
```

4 Convergence Analysis

4.1 Interval Reduction

At each iteration, the Bisection Method divides the current interval into two equal halves. Therefore, the interval length is reduced by a factor of two after every iteration.

Let the initial interval length be

$$L_0 = b - a.$$

Since the interval is halved at every iteration,

$$L_1 = \frac{L_0}{2},$$

$$L_2 = \frac{L_0}{2^2},$$

$$\vdots$$

Hence,

$$L_n = \frac{L_0}{2^n} = \frac{b-a}{2^n}.$$

4.2 Required Iterations

Suppose the desired accuracy is ϵ . Since the interval length after n iterations is

$$L_n = \frac{b-a}{2^n},$$

the stopping criterion requires

$$\frac{b-a}{2^n} \leq \epsilon.$$

Multiplying both sides by 2^n gives

$$b - a \leq \varepsilon 2^n.$$

Dividing both sides by ε ,

$$\frac{b - a}{\varepsilon} \leq 2^n.$$

Taking the base-2 logarithm of both sides,

$$\log_2 \left(\frac{b - a}{\varepsilon} \right) \leq n.$$

Therefore, the minimum number of iterations required is

$$n \geq \log_2 \left(\frac{b - a}{\varepsilon} \right)$$

This result shows that the number of iterations grows logarithmically with respect to the desired accuracy. Consequently, obtaining higher accuracy requires only a moderate increase in the number of iterations.

4.3 Error Bound

Let x^* denote the exact root and let m_n be the midpoint after the n th iteration.

Since the exact root always remains inside the current interval, the maximum error is bounded by half of the interval length:

$$|x^* - m_n| \leq \frac{L_n}{2}.$$

Substituting the interval length

$$L_n = \frac{b - a}{2^n},$$

gives

$$|x^* - m_n| \leq \frac{b - a}{2^{n+1}}.$$

Therefore, the approximation error decreases exponentially as the number of iterations increases.

4.4 Convergence Rate

The Bisection Method has **linear convergence**. The error is reduced by approximately one-half at each iteration. However, it is guaranteed to converge provided that

- the function is continuous on $[a, b]$, and
- $f(a)f(b) < 0$.

For this reason, the Bisection Method is regarded as one of the most robust root-finding algorithms.

5 Summary

The convergence properties of the Bisection Method can be summarized as follows:

- Interval length after n iterations:

$$L_n = \frac{b-a}{2^n}$$

- Required iterations:

$$n \geq \log_2 \left(\frac{b-a}{\varepsilon} \right)$$

- Error bound:

$$|x^* - m_n| \leq \frac{b-a}{2^{n+1}}$$

- Convergence order:

Linear Convergence

6 Numerical Example

To illustrate the Bisection Method, consider the nonlinear equation

$$f(x) = x^2 - 2.$$

The exact root is

$$x = \sqrt{2} \approx 1.41421356.$$

Choose the initial interval

$$[a, b] = [1, 2],$$

Since $f(1)f(2) < 0$, the Intermediate Value Theorem guarantees that at least one root exists in the interval $[1, 2]$.

Table 1: $f(x) = x^2 - 2$		
Iteration	m	$f(m)$
1	1.5000	0.2500
2	1.2500	-0.4375
3	1.3750	-0.1094
4	1.4375	0.0664
$\vdots$	$\vdots$	$\vdots$
27	1.41421	-5.23681×10^{-9}

After 27 iterations, the approximate root is 1.41421.